Townsend's 2026 Monthly Dust Loss Model

Purpose

This model is used to estimate the percentage of potential monthly photovoltaic (PV) energy sacrificed due to dust soiling.

PV financing decisions are based on long-term energy estimates, and popular PV modeling software such as PVsyst include monthly inputs for soiling losses – snow in the winter, dust in the summer. This modeling tool relies principally on monthly rainfall to determine a changing pattern of soiling throughout the year. That pattern is also influenced by four other factors:

1. The daily soiling ramp rate, entered as a single value for each season.
2. The number of manual array washings, up to two per year.
3. The influence of snow, where any month with at least a 3% projected snow loss will govern, resetting that month's dust soiling loss to zero.
4. For bifacial arrays, the back side's contribution is presumed to have no appreciable dust buildup, so the full array's monthly dust loss is reduced by the percentage of energy contributed from the back side.

Rules

The following table lists the core assumptions applied to each month, subject to the superseding influence of snow, which sets the dust loss to 0% if that month's snow losses are at least 3%.

Precipitation, inches (mm)	**Soiling increase from prior month**
≥4 (102)	0%
≥2 and <4 (51-102)	1%
≥1 and <2 (25-51)	If prior month precip. ≥2", add 8 days x seasonal ramp rate to prior month's soiling. If prior month precip. <2", add 15 days x seasonal ramp rate to prior month's soiling.
<1 (25)	Add product of number of days per month x seasonal ramp rate applicable to current month.

In addition to the above rules, several other assumptions are applied, too. The first concerns the matter of determining a starting month for a continuous but consistent pattern of monthly dust losses within a given year. The month with the most precipitation is chosen as the starting

month. The soiling applicable to that month is subject to the same rules described above, with one exception: if the starting month's precipitation is less than 2 inches, the soiling for that month is set to 2%.

A manual wash is assumed to occur on the first day of a month. Unlike the Kimber et al. model[i], this model does not include a "grace period" of 0% soiling after a manual wash. The dust loss in a manual wash month will still be calculated based on a half month of accumulation based on the ramp rate applicable to that season. If one or two manual washings per year are selected, the model does an exhaustive search, testing each month in sequence as the date of the first potential wash, recomputing the *annual* energy loss associated with each successive choice. After checking each month, the wash month with the lowest annual soiling loss is deemed to be optimal. The same logic is used to determine when to do a second wash. In almost all instances, the second wash, if chosen, will occur in a chronologically later month than the first wash, though there may be circumstances in which the combination of snow, rain, and the choice of the first wash may result in a second wash that is suggested for a month prior to the first wash. In practice, this should not either occur very often, if at all, and should be understood to mean that the value of the "second" wash is predicated on committing to perform the more impactful first wash in its prescribed month.

Inputs

1. Twelve monthly precipitation amounts, specified in units of inches or millimeters.
2. Four seasonal daily dust accumulation (ramp) rates, with four rate choices per season, with summer and fall periods tending to show higher ramp rates than in the spring or winter:
    a. 0.025%, for ultra-sandy, low moisture, and low particulate conditions
    b. 0.05%, desert
    c. 0.1%, typical default
    d. 0.15%, humid, agriculturally active, sooty, or prone to bird droppings
3. Number of manual washes, from 0 to 2 per year.
4. Bifacial: If bifacial, a separate calculation linked to the snow model is used to determine the monthly percentage of generation attributable to the essentially unsoiled back side, with the overall dust loss reduced accordingly.

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[i] Kimber, A., Mitchell, L., Nogradi, S. and Wenger, H. (2006) The Effect of Soiling on Large Grid-Connected Photovoltaic Systems in California and the Southwest Region of the United States. Proceedings of the 4th IEEE World Conference on Photovoltaic Energy Conference, Waikoloa, 2391-2395. https://doi.org/10.1109/WCPEC.2006.279690